

# KOUSHIK GADDAM

## AI / Software Engineer

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## SUMMARY

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Third-year CS (AI/ML) engineer specializing in deterministic, framework-agnostic AI-agent infrastructure and developer tooling. Builds production-grade systems end to end: AgentMesh, an agent observability and control plane with full CI/CD and Helm deployment, and CodeGraph, a 7-agent code-remediation swarm passing 96/96 tests with zero LLM-API dependency. Strengths in Go, Rust, and TypeScript across distributed backends, sandboxed execution, and provably correct systems. Seeking Summer 2026 SWE / Infrastructure / AI Engineering internships.

## EDUCATION

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### B.Tech, Computer Science Engineering (AI & ML) | Aug 2024 – Aug 2028

Keshav Memorial College of Engineering, Hyderabad, India

Relevant coursework: Data Structures & Algorithms, Operating Systems, DBMS, Distributed Systems, Machine Learning.

## PROJECTS

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### AgentMesh — Framework-agnostic AI-agent observability & control plane | Aug 2025

*Go, gRPC/Protobuf, ClickHouse, Postgres, Redis, MinIO, OpenTelemetry, Docker, Helm*

- Designed a schema-first telemetry data model over ClickHouse, Postgres, Redis, and MinIO to capture full-fidelity agent traces [Add throughput / agents observed].
- Built OpenTelemetry-native adapters unifying LangGraph, CrewAI, AutoGen, and the OpenAI Agents SDK behind a single integration surface.
- Engineered a deterministic replay engine that reproduces historical agent failures exactly, enabling postmortem debugging without live reproduction.
- Shipped an MCP governance layer with policy enforcement, capability scoping, and audit trails, delivered with full CI/CD, Docker, and Helm/Kubernetes deployment [Add GitHub stars / contributors].

### CodeGraph — Deterministic multi-agent code remediation | Oct 2025

*Rust, Go, TypeScript, TS Compiler API, VS Code API, Next.js*

- Modeled arbitrary git repositories as traversable symbol graphs; implemented Tarjan SCC + BFS for blast-radius-weighted health scoring [Add LOC / repo size].
- Built a deterministic 7-agent pipeline (sandbox to codemod to re-index to verified diff) requiring no LLM API key, with all 96 suite tests passing.
- Engineered a verify-before-apply guard that validates fixes in a disposable sandbox and emits a diff only if the health score does not regress [Add real bugs fixed].
- Hardened the system from 3/10 to 8/10 through an open self-audit and a 99-issue security review with published results.
- Shipped as a hosted Next.js app and a standalone offline VS Code extension [Add GitHub stars / installs].

### BrowseMesh — Offline, private browsing knowledge engine | Nov 2025

*TypeScript, IndexedDB, Vector Search*

- Built an on-device entity-extraction and embedding pipeline powering hybrid full-text and vector (RAG-style) search across browsing history, fully offline with no server [Add latency / corpus size].
- Developed a plugin SDK over a typed IndexedDB data tier for third-party extensibility.

## SuperSearch — Capability-secured natural-language command runner | Jan 2026

*Rust, WASM (wasmtime), Tauri, React, Framer Motion*

- Built a compiler that turns natural language into replayable, inspectable execution graphs, guarded by an object-capability model and an append-only journal.
- Engineered a wasmtime plugin host with fuel and memory limits, sandboxing third-party extensions under deterministic resource budgets.
- Implemented an argv-only dispatch path with per-call capability gating, eliminating shell injection and ambient authority.
- Crafted a Spotlight-grade command palette in React and Framer Motion over Tauri IPC for a native macOS experience.

## Live API Inspector — Privacy-first API DevTools extension | Mar 2026

*TypeScript, React 19, Vite, Manifest V3*

- Built a zero-config Manifest V3 DevTools extension performing semantic endpoint clustering, latency analysis, and on-device secret redaction, keeping all request data in the browser [Add installs].

## SKILLS

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**Programming Languages:** Go, Rust, Python, TypeScript

**Backend:** gRPC/Protobuf, REST/API Design, WASM (wasmtime), Tauri, Concurrency

**Frontend:** React, Next.js, Vite, Tailwind CSS, Framer Motion, Manifest V3

**Databases:** ClickHouse, Postgres, Redis, MinIO, IndexedDB

**Cloud:** Render, MinIO (S3-compatible)

**DevOps:** Docker, Helm, Kubernetes, CI/CD, Git

**AI / ML:** LangGraph, CrewAI, AutoGen, OpenAI Agents SDK, Model Context Protocol (MCP), OpenTelemetry, Vector Search, RAG, Deterministic Replay

**Infrastructure:** OpenTelemetry, Helm/Kubernetes, Docker, Distributed Data Stores

**Security:** Object-Capability Model, Sandbox Execution, Adversarial Testing, Static Analysis, Append-Only Journals, Open Security Audits

**Developer Tools:** VS Code API, TS Compiler API, Vite, Manifest V3

## OPEN SOURCE

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Public repositories ([github.com/archdex-art](https://github.com/archdex-art)): AgentMesh, CodeGraph, SuperSearch, BrowseMesh, Live API Inspector. [Add stars / external contributors / merged PRs.]